

1. Types of Shock

- Hypovolemic Shock
- Cardiogenic Shock
- Distributive Shock: Septic, Anaphylactic, Neurogenic
- Obstructive Shock

3. VAP Bundle

- Head of bed (HOB) elevation 30–45°
- Daily sedation vacations + readiness to wean assessment
- Peptic ulcer disease (PUD) prophylaxis
- Deep venous thrombosis (DVT) prophylaxis
- Daily oral care with chlorhexidine

5. Causes of Respiratory Failure

- Airway obstruction
- Central nervous system (CNS) depression
- Neuromuscular diseases: Myasthenia gravis (MG), Guillain-Barré syndrome (GBS)
- Chest wall and pleural deformities
- Pulmonary diseases: Pneumonia, Chronic obstructive pulmonary disease (COPD), Pulmonary edema, Acute respiratory distress syndrome (ARDS)

7. Acute Coronary Syndrome (ACS) Types by Occlusion

- ST-elevation myocardial infarction (STEMI): Complete occlusion
- Non-ST-elevation myocardial infarction (NSTEMI): Partial occlusion with myocardial damage
- Unstable Angina (UA): Partial occlusion without myocardial damage

9. Nursing Interventions to Decrease Intracranial Pressure (ICP)

- Keep head midline and elevate head of bed (HOB) to 30°
- Avoid hip flexion and neck rotation
- Space out nursing activities
- Minimize suctioning (pre-oxygenate if needed)
- Maintain a quiet and calm environment

2. 5 ICU Vital Signs (besides BP, Pulse, RR, Temp, Pain)

- Glasgow Coma Scale (GCS): Normal = 15
- Random Blood Sugar (RBS): Fasting 70–120 mg/dL, Postprandial (PP) < 180 mg/dL
- O₂ Saturation: Normal > 96%
- End-Tidal CO₂ (EtCO₂): Normal 35–45 mmHg
- Central Venous Pressure (CVP): Normal 6–8 mmHg

4. Indications for Mechanical Ventilation (MV)

- Apnea or absence of breathing
- Acute ventilatory failure: pH ≤ 7.25 with PaCO₂ ≥ 50 mmHg
- Severe hypoxemia: PaO₂ ≤ 50 mmHg on room air
- Respiratory muscle fatigue

6. Glasgow Coma Scale (GCS)

- Eye Opening (E): Spontaneous (4), To speech (3), To pain (2), None (1)
- Verbal Response (V): Oriented (5), Confused (4), Inappropriate words (3), Incomprehensible (2), None (1)
- Motor Response (M): Obeys (6), Localizes (5), Withdrawal (4), Abnormal flexion (3), Extension (2), None (1)
- Total Score: 3 (Deep Coma) → 15 (Normal)

8. Vagotonic Maneuvers to Decrease Heart Rate (HR)

- Carotid sinus massage
- Valsalva maneuver
- Facial immersion in cold water
- Ocular pressure
- Self-induced vomiting
- Breath holding
- Head lowered between the knees

10. Complications of Diabetic Ketoacidosis (DKA)

- Cerebral edema
- Hypoglycemia (treatment-related)
- Hypokalemia (treatment-related)
- Aspiration
- Acute respiratory distress syndrome (ARDS)

11. Treatment of Diabetic Ketoacidosis (DKA)

- Fluid replacement: Normal saline (NS)
- Electrolyte replacement: Potassium (K+) monitoring and replacement
- Insulin therapy: Continuous IV insulin infusion
- Correct acidosis: Sodium bicarbonate (NaHCO₃) only if pH < 7.0

12. Symptoms and Signs of DKA

- Symptoms: Polyuria, polydipsia, polyphagia, nausea, vomiting, abdominal pain
- Signs: Kussmaul respirations, fruity (acetone) breath odor
- Signs: Dehydration — hypotension, tachycardia
- Signs: Altered mental status (AMS)

13. Stroke Types and Target Blood Pressure (BP)

- Ischemic stroke: Treat only if BP > 220/120 mmHg
- Ischemic (thrombolysis candidate): Target BP < 185/110 mmHg before treatment
- Hemorrhagic stroke: Target systolic blood pressure (SBP) 140–160 mmHg

14. Disseminated Intravascular Coagulation (DIC) Diagnosis

- Clinical: Bleeding from multiple sites (petechiae, ecchymosis, IV sites)
- Clinical: Signs of microvascular thrombosis leading to organ failure
- Labs: Prolonged prothrombin time (PT) / partial thromboplastin time (PTT)
- Labs: Decreased platelets, decreased fibrinogen, elevated D-dimer

15. Causes of Upper Gastrointestinal (GI) Bleeding

- Peptic ulceration
- Erosion / mucosal inflammation: Esophagitis, gastritis
- Esophageal or gastric varices
- Mallory-Weiss tear
- Tumors

16. Precipitants of Hepatic Encephalopathy (HE)

- Gastrointestinal (GI) bleeding
- Infections: Spontaneous bacterial peritonitis (SBP)
- Constipation
- Dehydration or electrolyte imbalance
- High protein intake
- Use of sedatives

17. Nursing Interventions for Hepatic Encephalopathy (HE)

- Administer lactulose to promote ammonia excretion
- Administer antibiotics: Rifaximin
- Maintain low-protein diet during acute phases
- Monitor neurological status frequently

18. Target Organ Damage in Hypertensive Emergency

- Central nervous system (CNS): Encephalopathy or intracranial hemorrhage (ICH)
- Eyes: Grade 3–4 retinopathy or papilledema
- Heart: Congestive heart failure (CHF), myocardial infarction (MI)
- Kidneys: Acute kidney injury (AKI) or hematuria
- Vasculature: Aortic dissection

19. Causes of Acute Respiratory Distress Syndrome (ARDS)

- Direct lung injury: Pneumonia, gastric aspiration, lung contusion, drowning
- Indirect lung injury: Sepsis, severe trauma, massive blood transfusion, acute pancreatitis

20. Complications of Mechanical Ventilation (MV)

- Ventilator-associated pneumonia (VAP)
- Barotrauma / volutrauma (e.g., pneumothorax)
- Decreased cardiac output (CO) from high intrathoracic pressure
- Stress ulcers in the gastrointestinal (GI) tract

21. Mechanical Ventilation (MV) Alarms

- High pressure: Coughing, secretions, or biting the tube
- Low pressure: Disconnection or leaks
- Low exhaled volume
- Apnea alarm

22. Eclampsia

- Definition: Severe hypertension during pregnancy with seizures
- Treatment: Volume expansion
- Treatment: Magnesium sulfate (MgSO₄) for seizure prophylaxis and treatment
- Treatment: Antihypertensives — Hydralazine

23. Management of Acute Respiratory Failure

- Treat the underlying cause
- Provide oxygen (O₂) therapy and maintain airway
- Restore fluid and electrolyte balance
- Optimize cardiac function and cardiac output (CO)

24. Complications of Acute Respiratory Failure

- Ischemic anoxic encephalopathy
- Cardiac dysrhythmias
- Venous thromboembolism (VTE)
- Gastrointestinal (GI) bleeding

25. Differential Diagnosis of Stroke

- Hypoglycemia
- Seizures / Post-ictal state
- Brain tumor or abscess

26. Prevention of Stroke

- Control hypertension (HTN), diabetes mellitus (DM), and hyperlipidemia
- Lifestyle: Smoking cessation, healthy diet, regular exercise
- Medication: Antiplatelets or anticoagulants for high-risk patients

- Meningitis or encephalitis

27. Management of Stroke

- Ischemic: Thrombolytics — recombinant tissue plasminogen activator (rtPA) if within time window
- Ischemic: Antiplatelets — Aspirin; supportive blood pressure (BP) and glucose control
- Hemorrhagic: Blood pressure (BP) control and intracranial pressure (ICP) management
- Hemorrhagic: Potential surgical evacuation